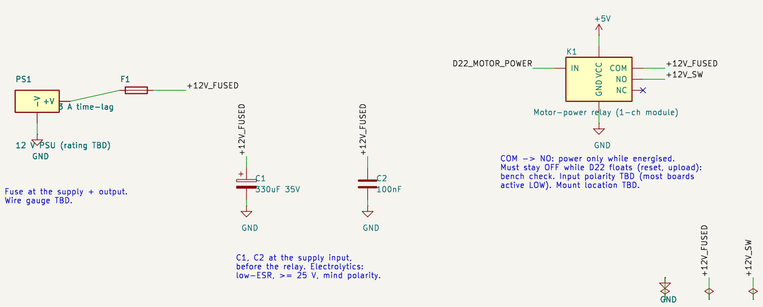
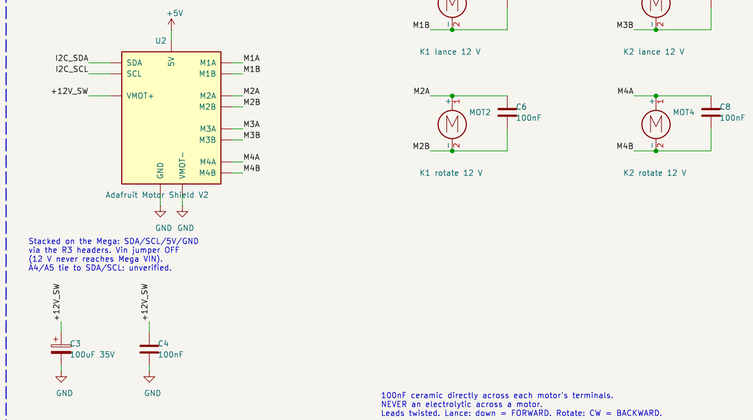
JOUST v2 — WIRING SCHEMATIC

Pins: firmware/include/config.h (source of truth). Same-name labels are connected. TBD = owner to confirm. Block diagram + pin table: docs/wiring/README.md

12 V MOTOR POWER (single supply, one fuse, one cut-off relay)

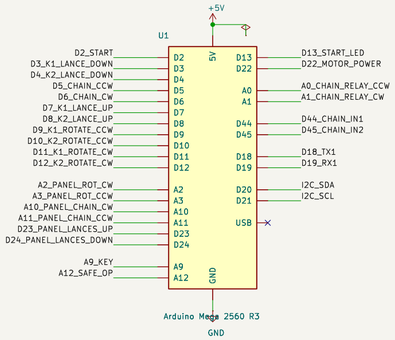


MOTOR SHIELD + KNIGHT MOTORS

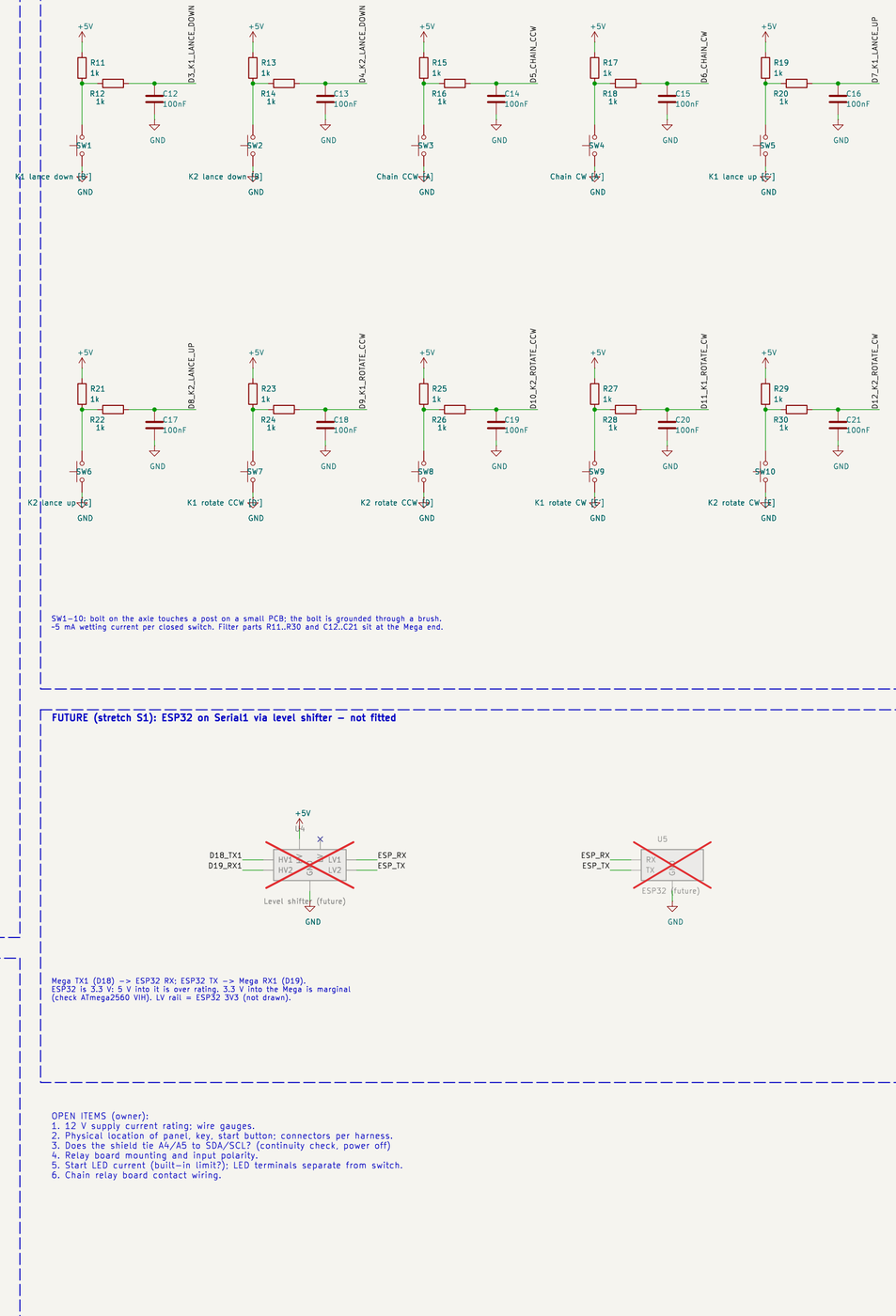


CONTROLLER (Mega on USB; shield stacked on top)

All inputs: switch to GND,
active LOW, pull-up.

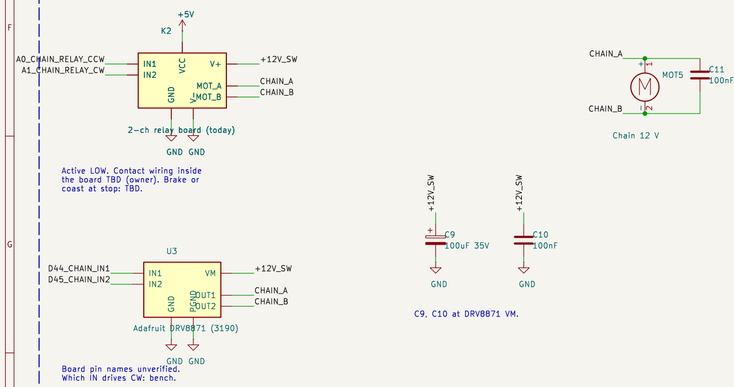


LIMIT INPUT FILTERS (protoboard near the Mega), x10: 1k pull-up to 5 V, 1k series, 100nF at the pin

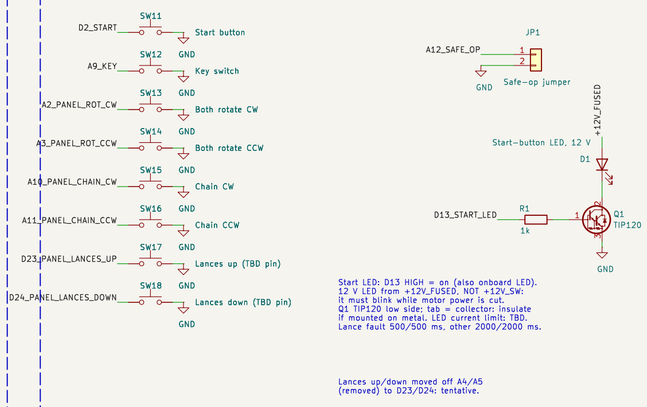


FUTURE (stretch S1): ESP32 on Serial1 via level shifter — not fitted

CHAIN DRIVE: relay board today -> DRV8871 at Task 9 (only one fitted)



OPERATOR INPUTS + INDICATOR (internal pull-ups, no filter)



OPEN ITEMS (owner):
1. 12 V supply current rating; wire gauges.
2. Physical location of panel, key, start button; connectors per harness.
3. Does the shield tie A4/A5 to SDA/SCL? (continuity check, power off)
4. Relay board mounting and input polarity.
5. Start LED current (built-in limit?); LED terminals separate from switch.
6. Chain relay board contact wiring.

Pins: firmware/include/config.h. Draft for owner review.
Generated by docs/wiring/tools/gen_schematic.py — edit in KiCad from here on
Miniature World — Camelot

Sheet: /
File: joust-v2.kicad_sch

Title: Joust v2 — wiring

Size: A2	Date: 2026-11-08	**Rev: 0.1**
KiCad E.D.A. 9.0.8		Id: 1/1